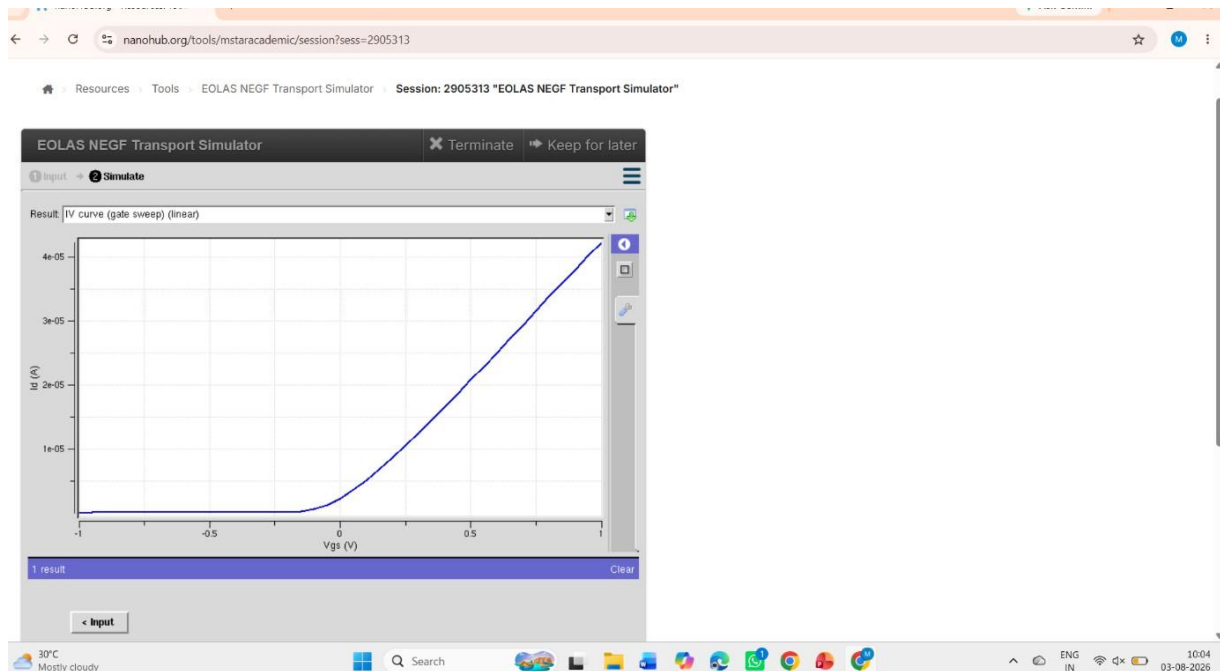


Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using Nano Hub

(a) FINFET

Output: The output Drain current Performance Vs Gate supply voltage of FinFET device



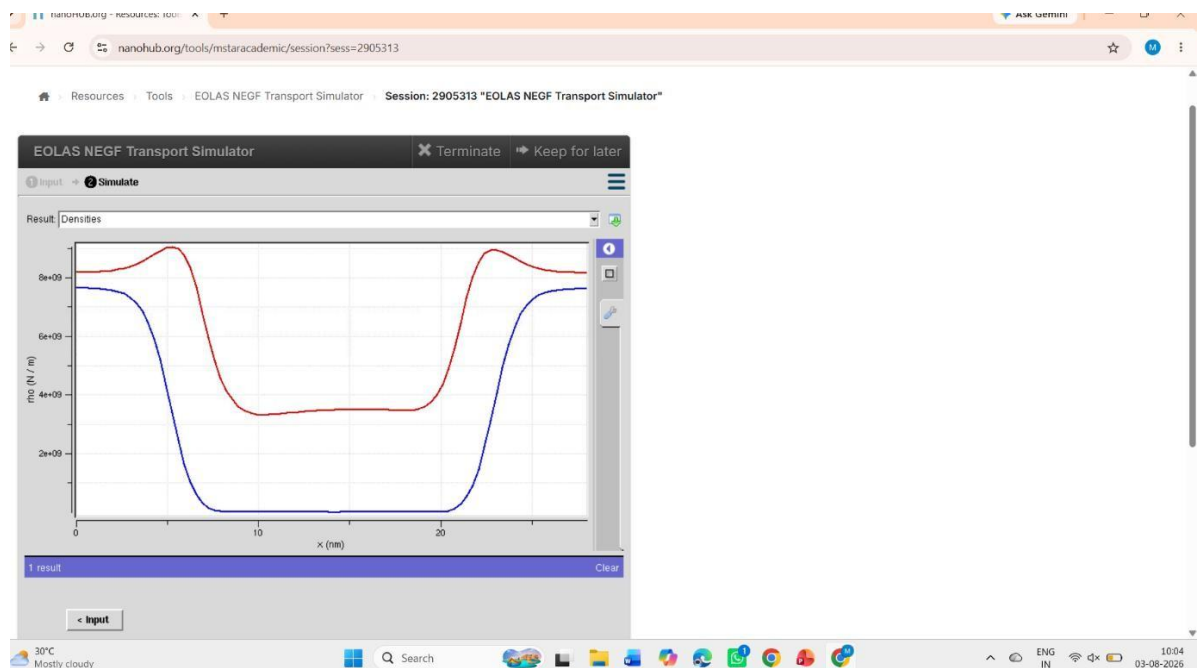
Outcome: *The drain current (I_d) increases rapidly with increasing gate voltage (V_{gs}), demonstrating effective gate-controlled switching behavior.*

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET . Analyze the performance for different Oxide layer thicknesses



Outcome: *The energy-resolved current is highest near the Fermi level and decreases with increasing energy, indicating carrier transport is concentrated at lower energies.*

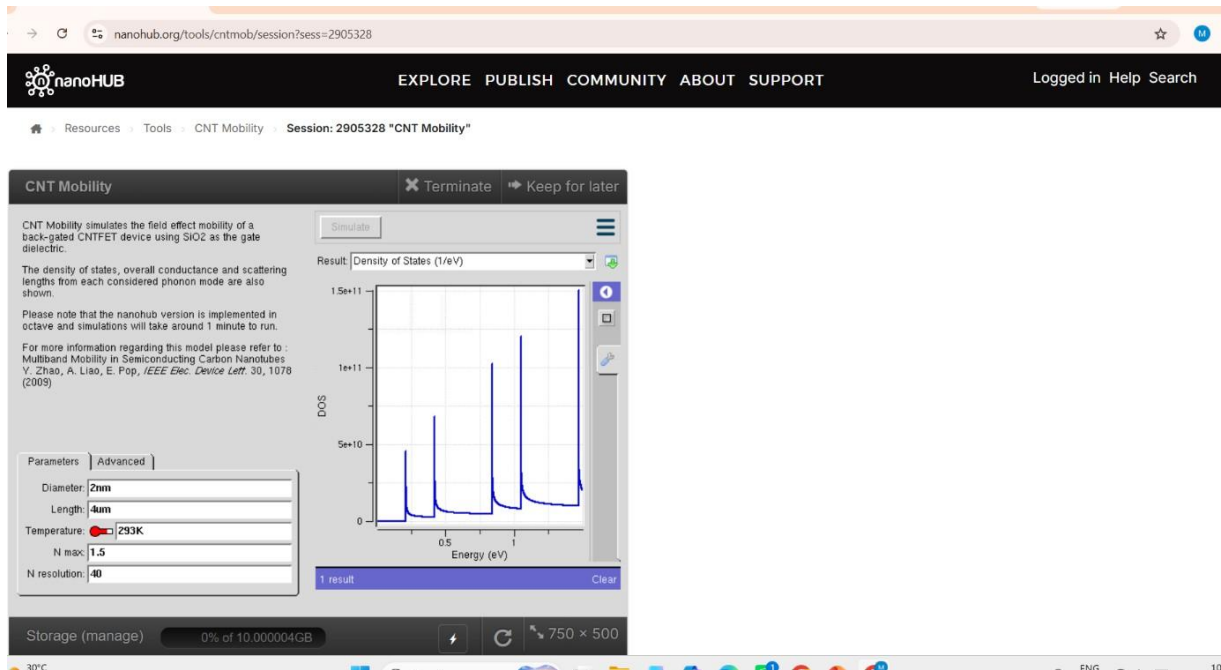
Case Study 2: Simulate & obtain the electron density per sheet characteristics of FinFET. Analyze its performance with different Gate region length



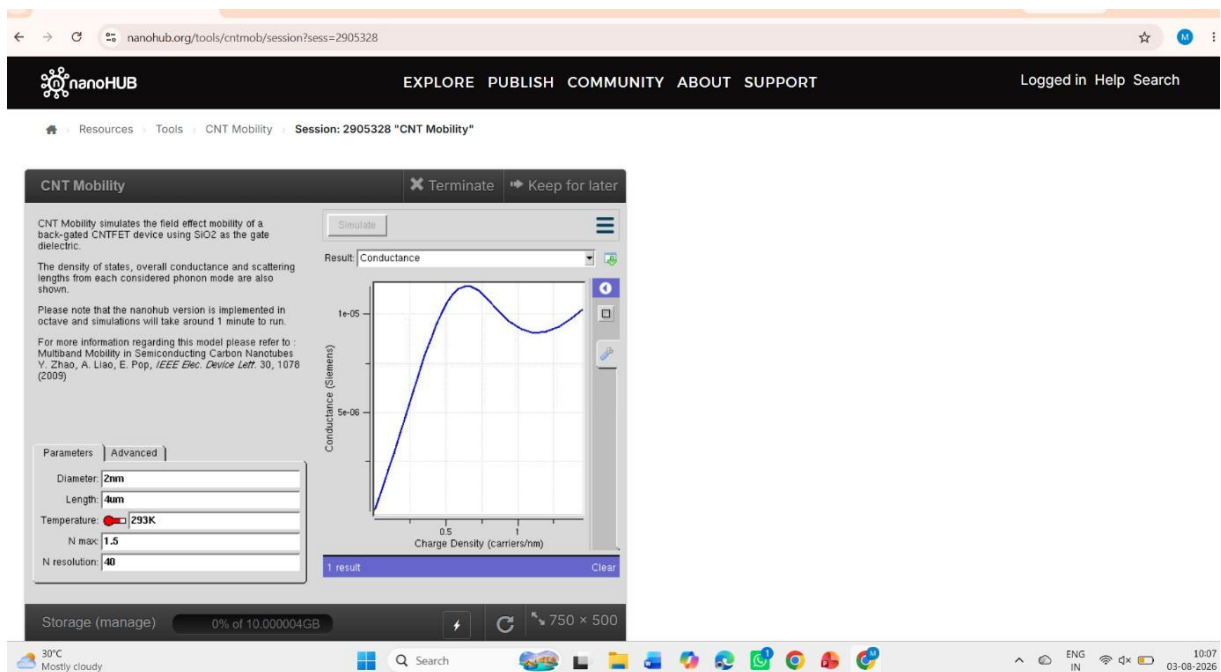
Outcome: *The electron and hole densities vary across the device, showing carrier accumulation near the contacts and reduced density in the channel*

(b) CNTFET

Output:

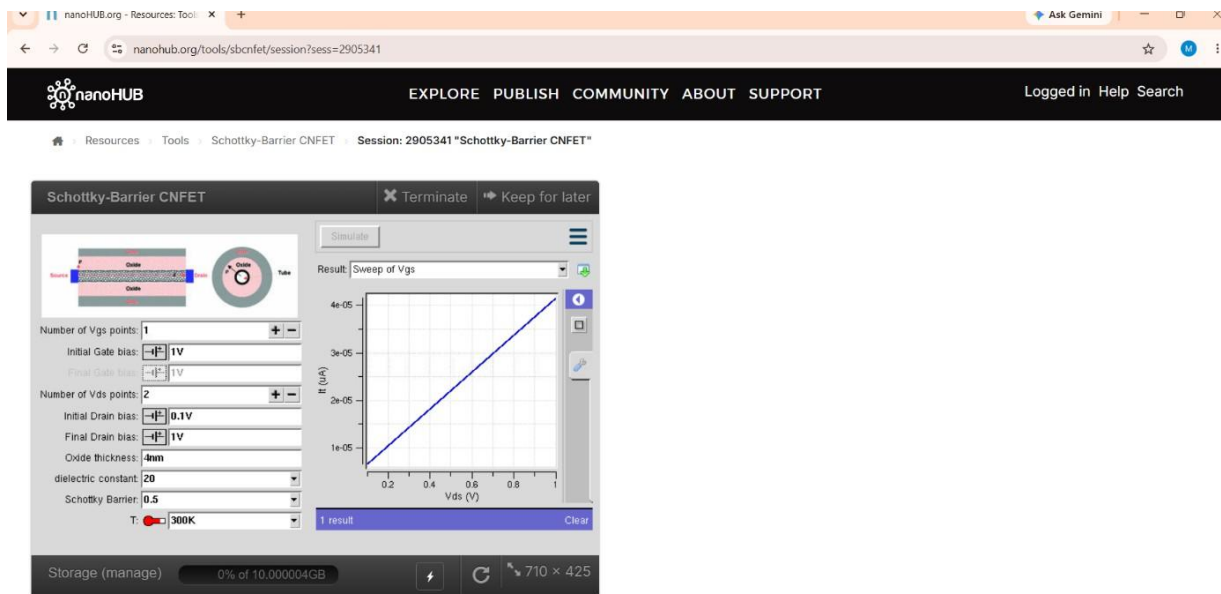


Outcome: *The density of states exhibits sharp peaks with increasing energy, indicating quantized energy levels in the carbon nanotube.*



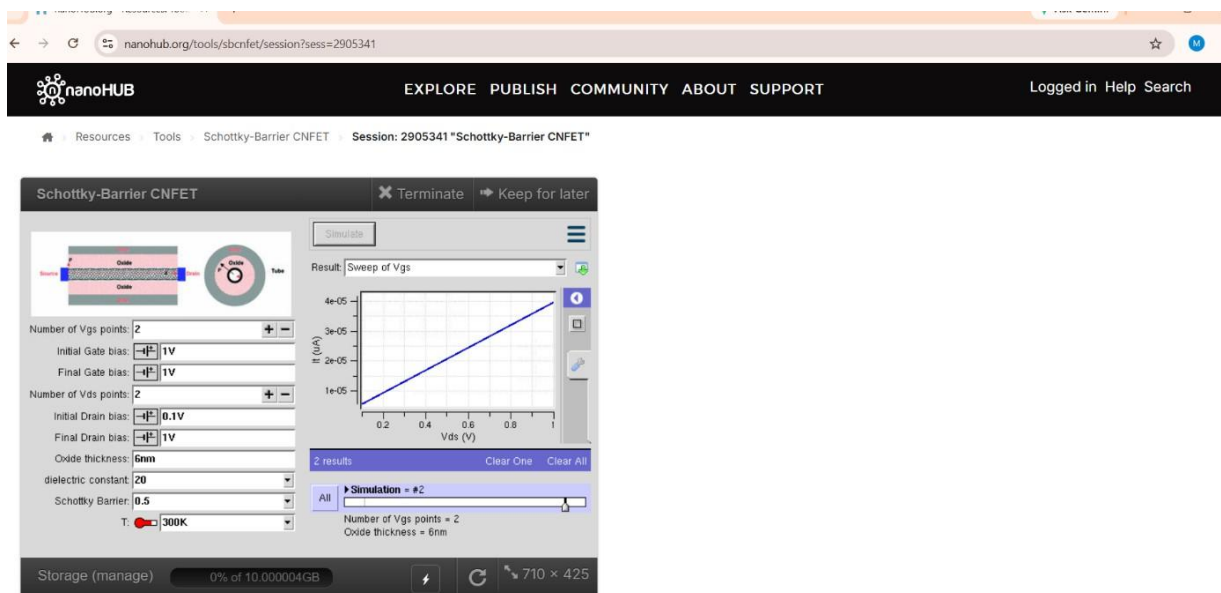
Outcome: *The conductance increases with carrier density, showing improved electrical transport as more charge carriers are available.*

Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



Outcome: The drain current increases nearly linearly with drain voltage, indicating stable current transport through the Schottky-barrier CNFET.

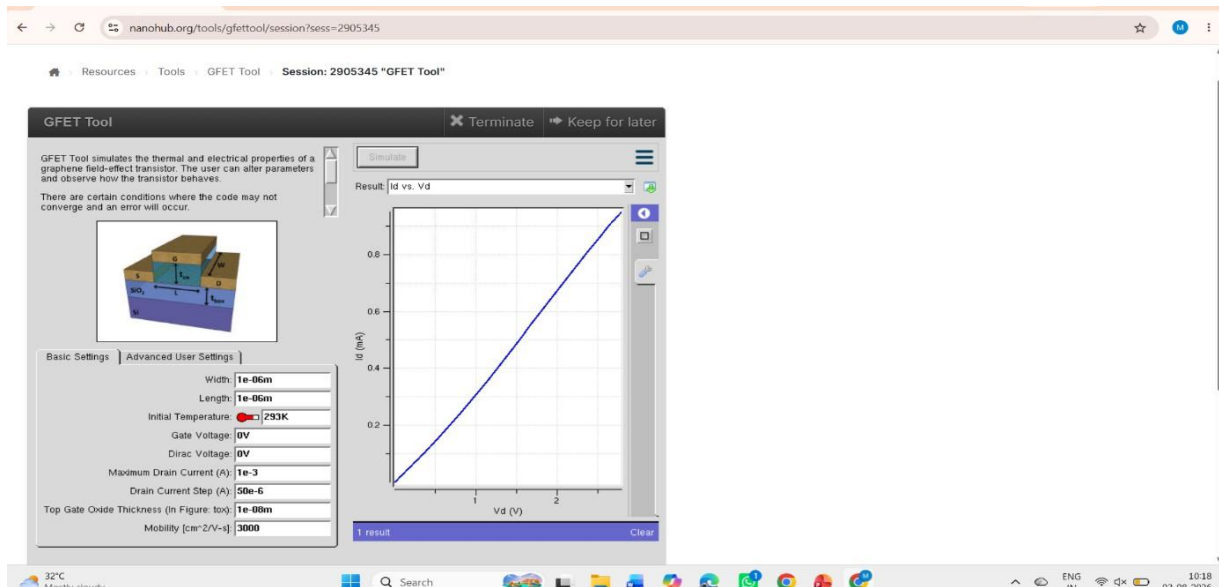
Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter



Outcome: The drain current increases with applied drain voltage under different gate bias conditions, confirming effective transistor operation.

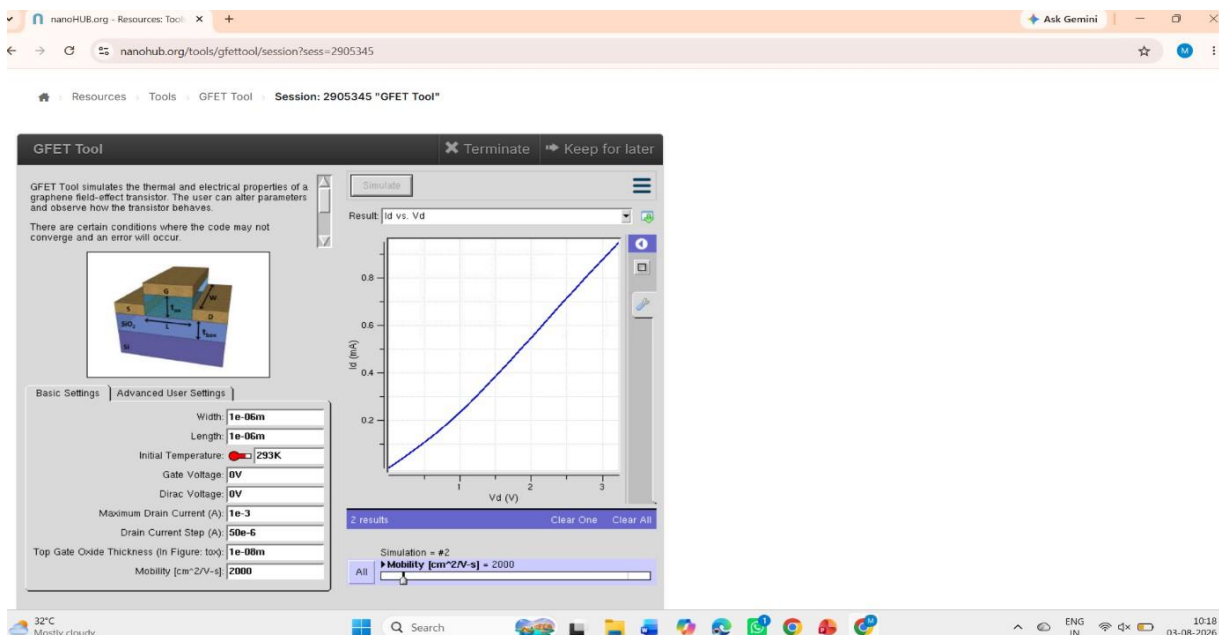
(c) GFET

Case Study 1: How does varying the gate voltage (V_{gs}) affect the drain current (I_{ds}) in a GFET?



Outcome: The drain current (I_d) increases almost linearly with drain voltage (V_d), indicating good current conduction in the GFET.

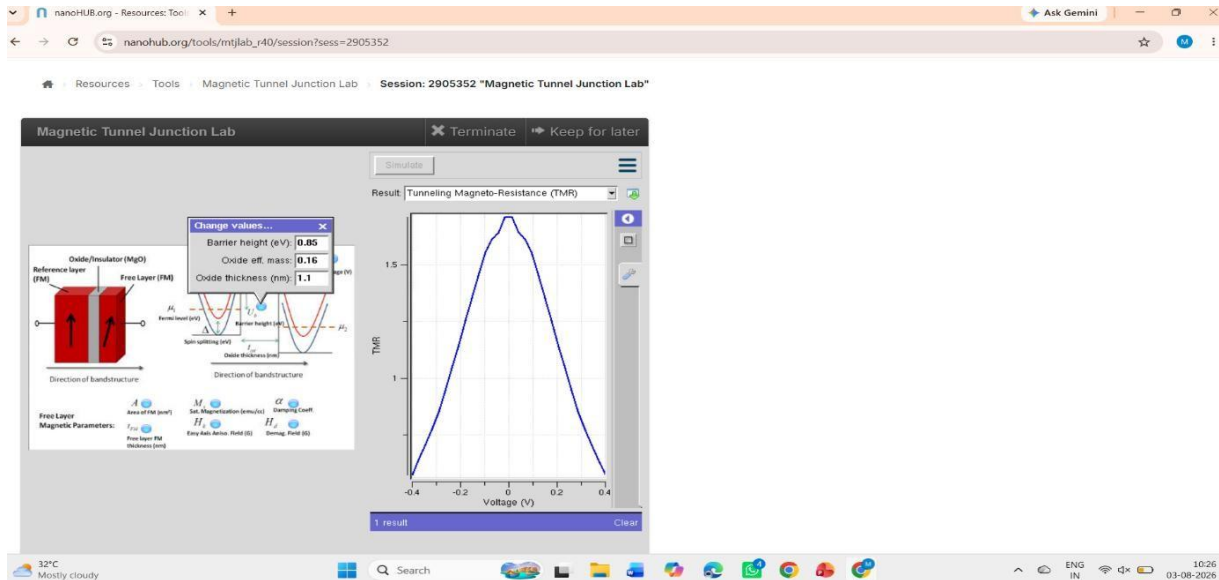
Case Study 2: What factors determine the maximum drain current ($I_{ds,max}$) in a Graphene Field Effect Transistor (GFET)



The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel

Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using Nano HUB

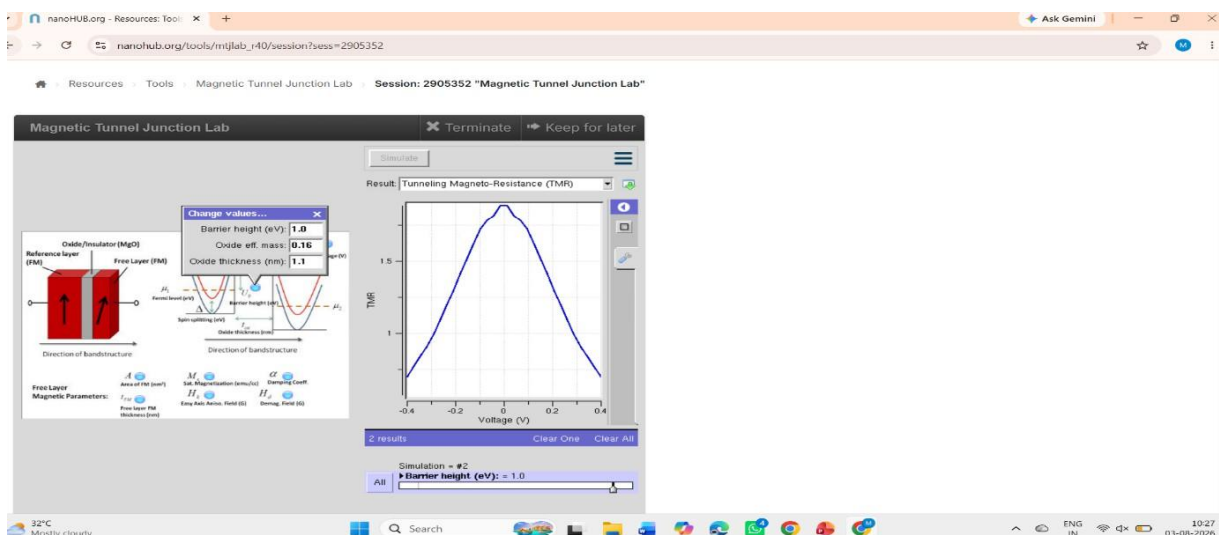
Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Output : TMR vs Voltage (Barrier Height = 0.85 eV):

"The TMR reaches its maximum at 0 V and decreases symmetrically as the applied voltage increases in either direction."

Case Study 2: Characteristics of TMR with barrier height of 1.0eV

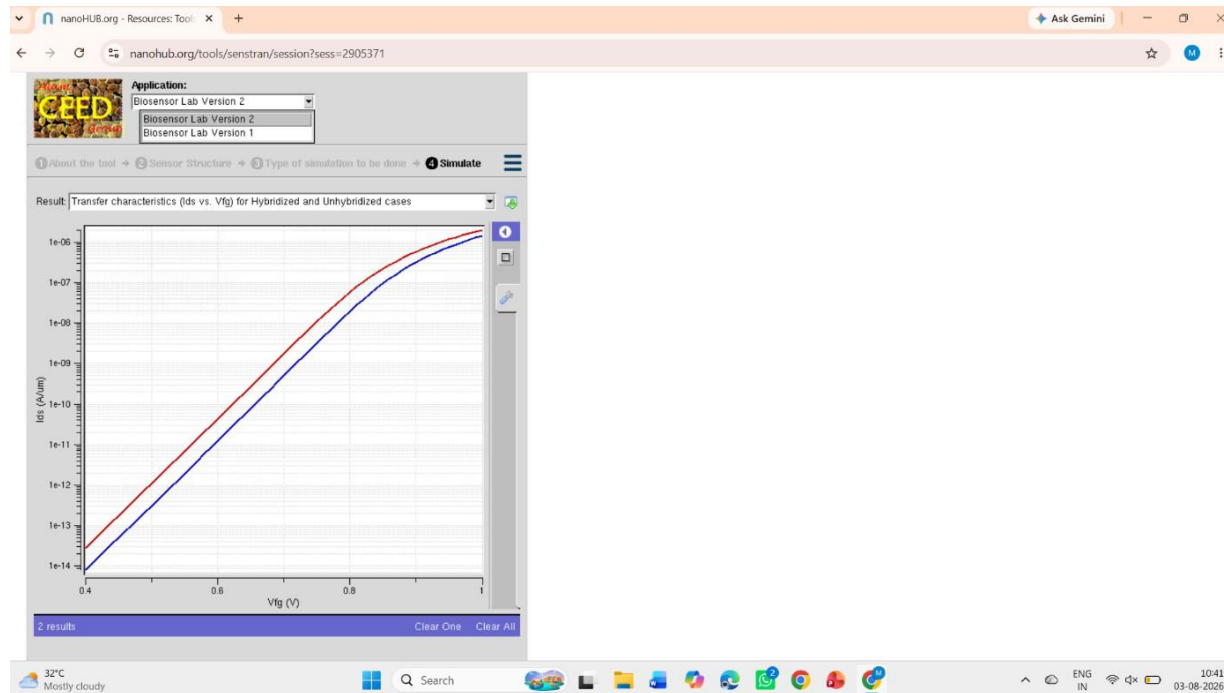


Output: TMR vs Voltage (Barrier Height = 1.0 eV):

"The TMR exhibits a peak near 0 V and decreases symmetrically with increasing positive or negative voltage, indicating optimal performance at zero bias."

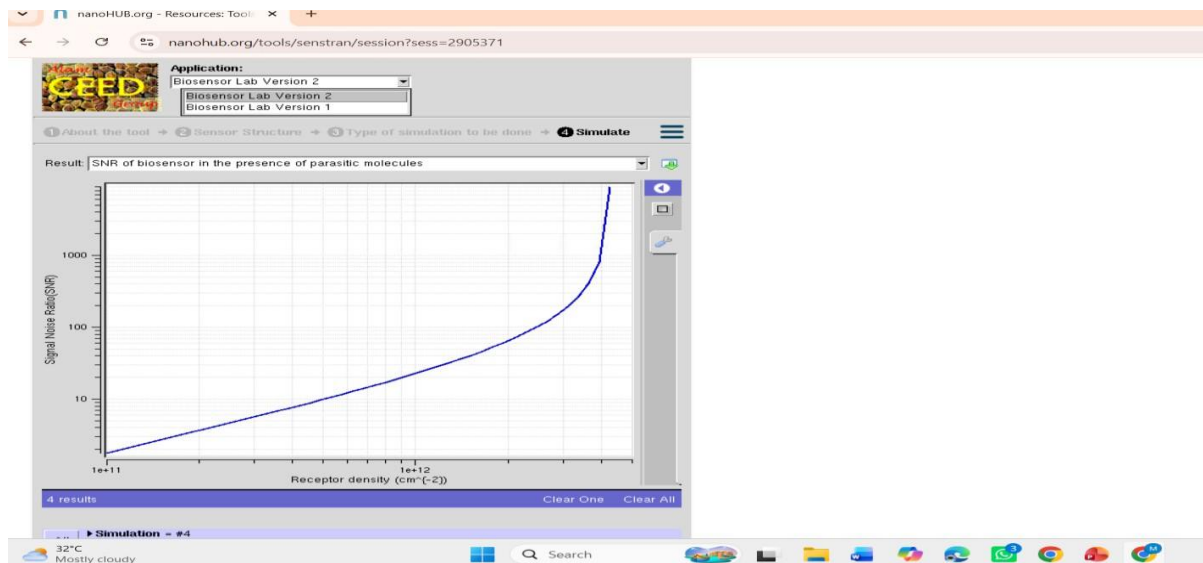
Exp. 10. Biosensor and Bio-FET characteristics using NanoHub

Case Study 1: *Comparative Analysis of Hybridized and Unhybridized BioFET Transfer Characteristics for Enhanced Biosensing Performance*



Outcome: On an industrial scale the film deposition is done by magnetron sputter deposition; on a laboratory scale molecular beam epitaxy, pulsed laser deposition and electron beam physical vapor deposition are also utilized. The junctions are prepared by photolithography.

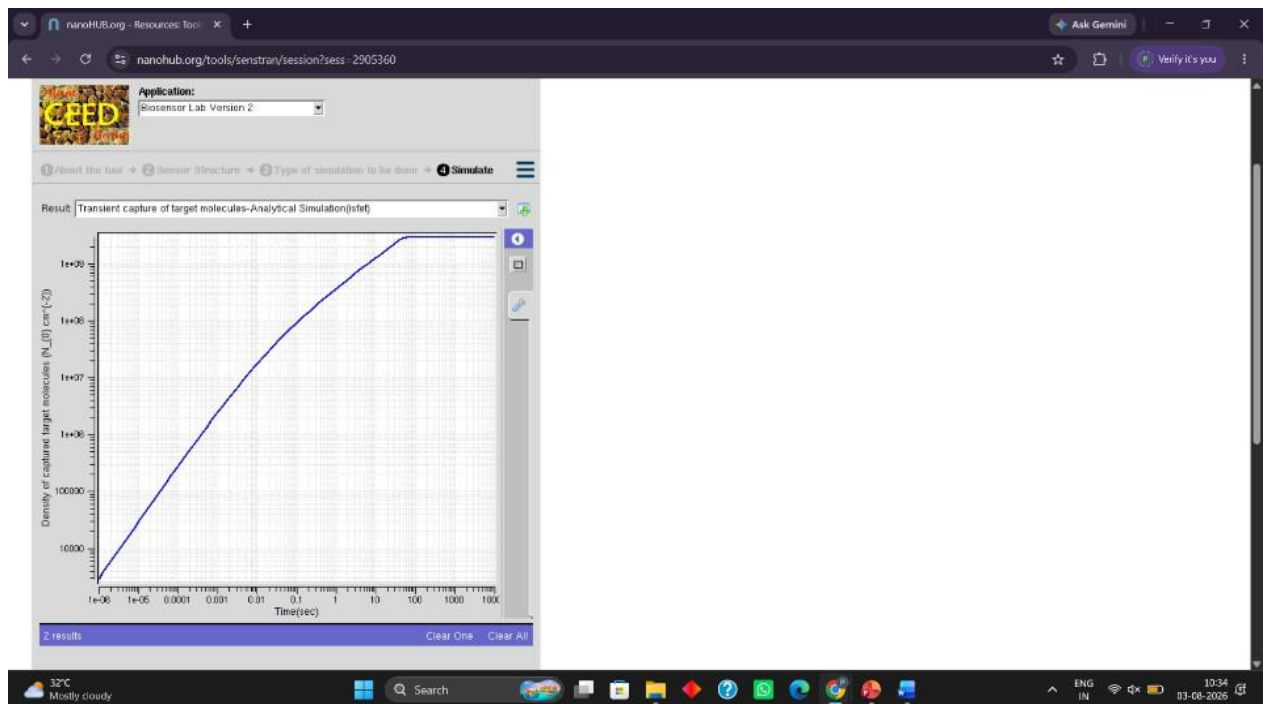
Case Study 2: *Evaluation of Signal-to-Noise Ratio (SNR) Variation with Receptor Density in the Presence of Parasitic Molecules*



Outcome: The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source

voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ($2000 \text{ cm}^2/\text{Vs}$) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

Case Study 3: *Transient Analysis of Target Molecule Capture Dynamics in a BioFET-Based Biosensor*



Outcome: The simulated VI curves show a relatively high drain current (I_{ds}) for a given gate-source voltage (V_{gs}), demonstrating efficient carrier transport in the graphene channel. High carrier mobility ($2000 \text{ cm}^2/\text{Vs}$) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.